

AMD Versal™ RF Series

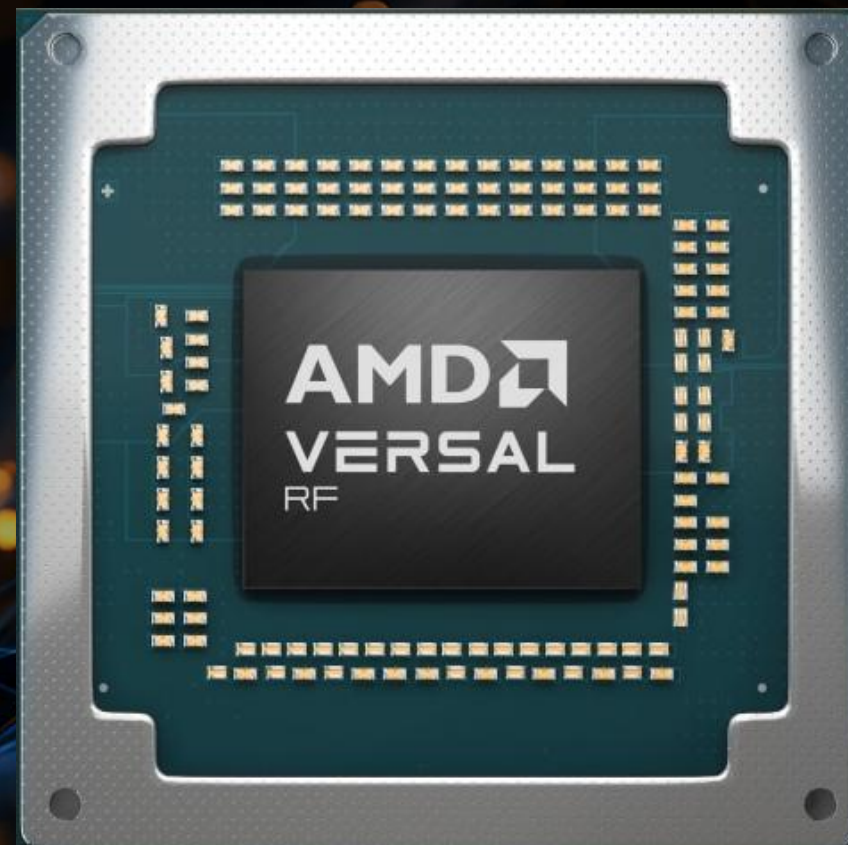
Jeff Cuppett

Technical Director, Chip Lead

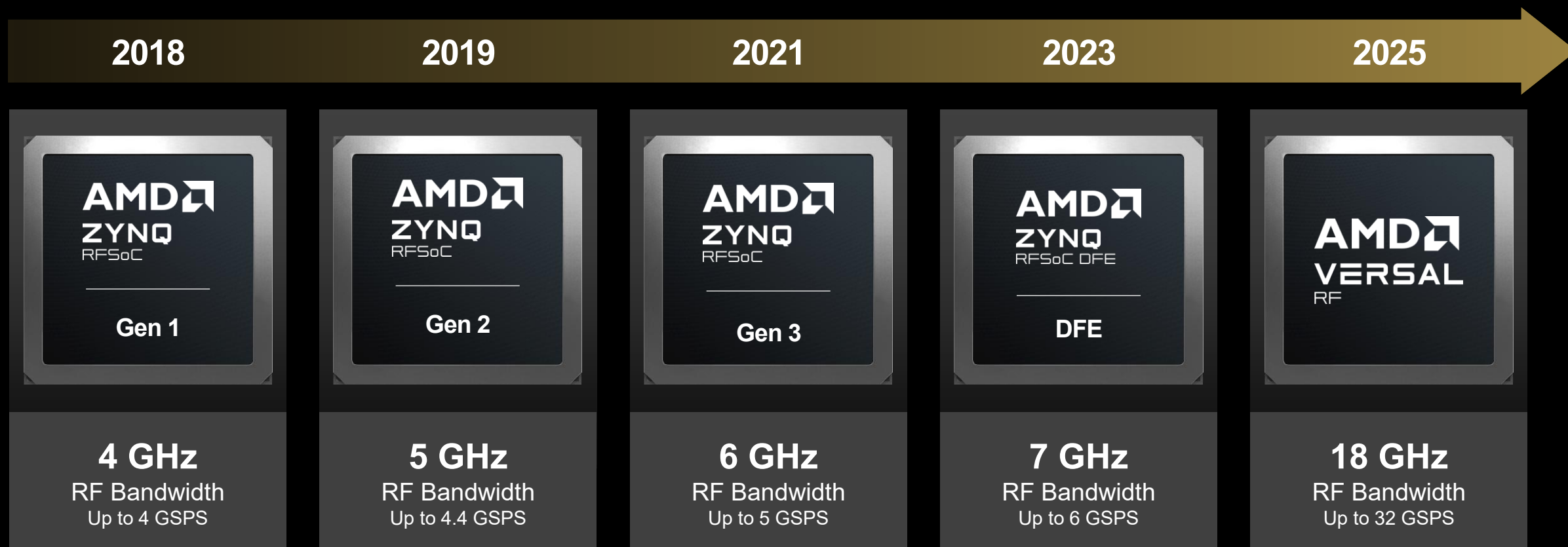
Martin Lee, PhD

Director, Product Planning

Hot Chips 2026



Continuing Leadership in Integrated RF

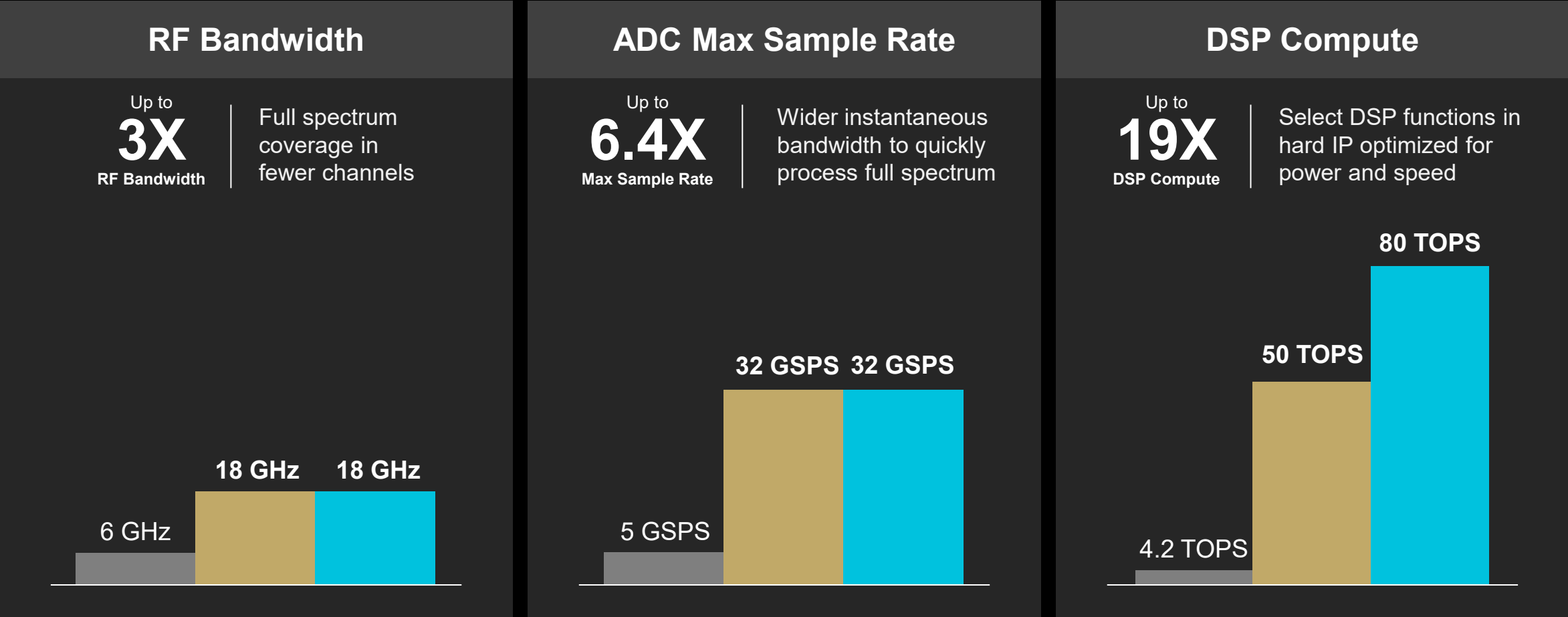


Compute and RF Performance vs. Previous Generation

AMD Zynq™ RFSoc Gen 3
Adaptive SoC

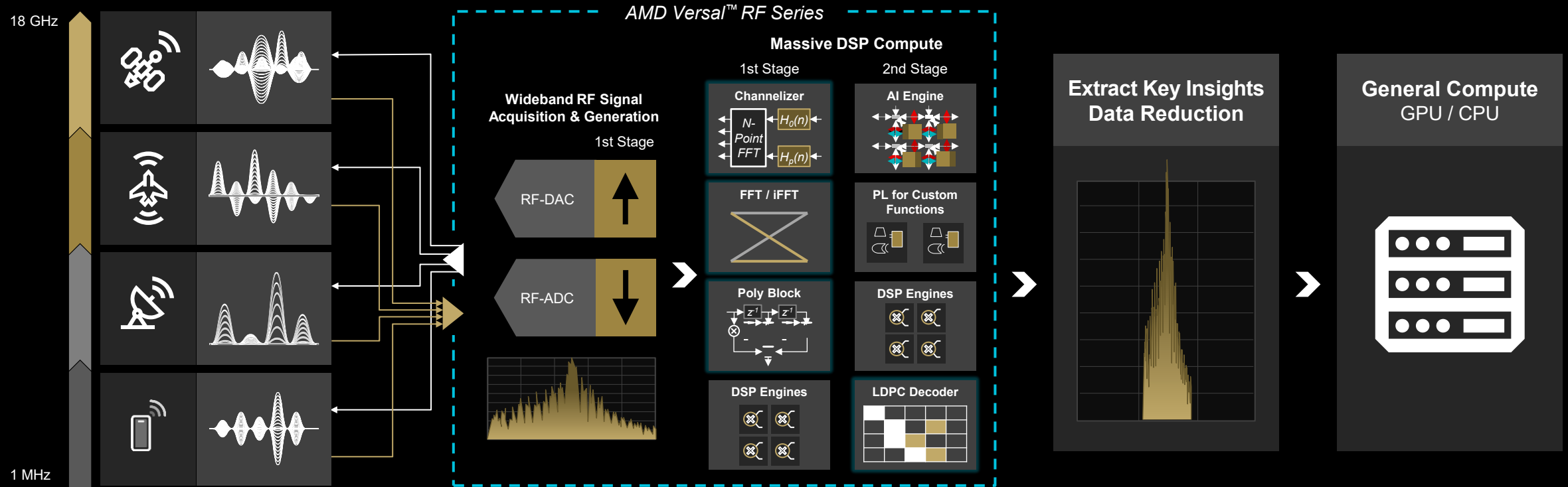
AMD Versal™ RF Series
VR1602/1652 Devices

AMD Versal™ RF Series
VR1902/1952 Devices

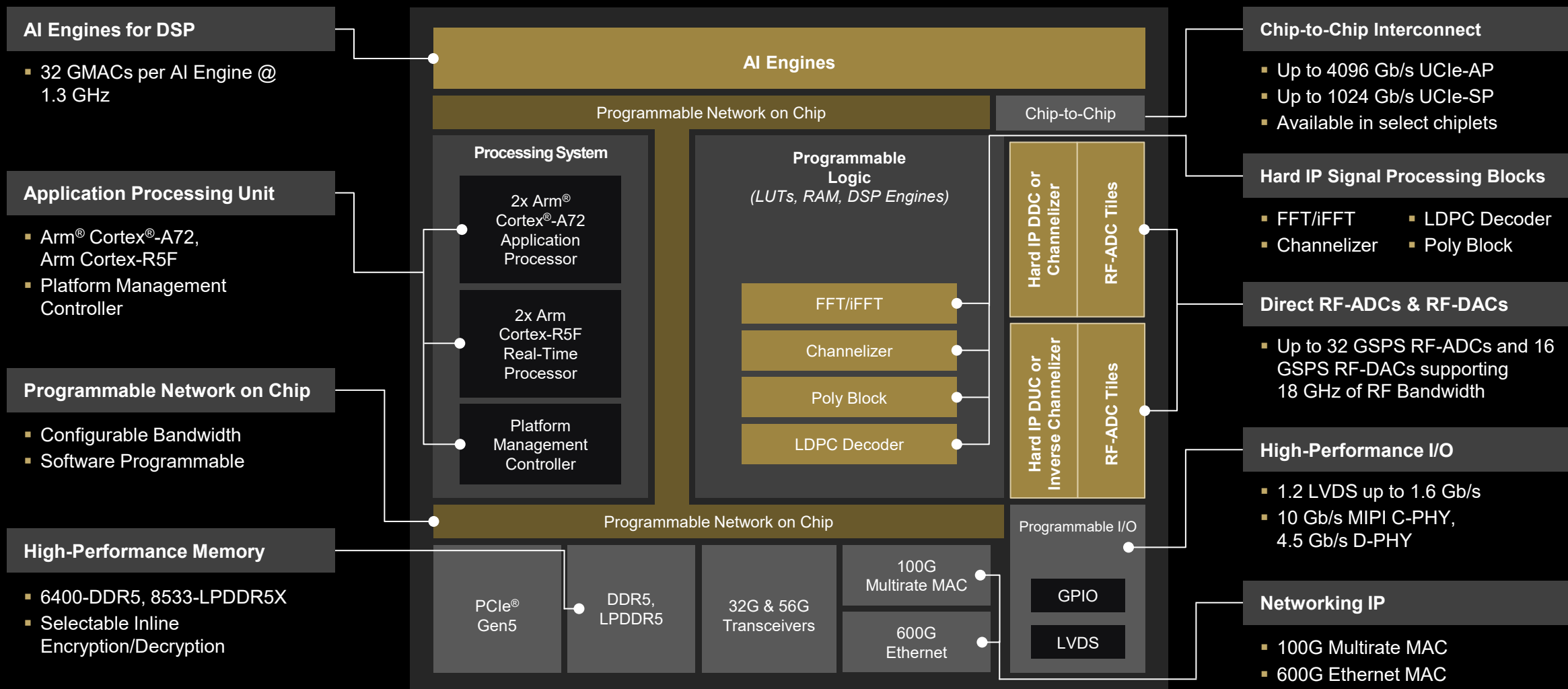


Problem Statement - Drinking from the RF Firehose

- Wideband RF and high sample rates require **massive DSP compute**
- RF front-end processing must acquire and process wideband RF signals with **low latency**
- Far-edge RF processing needs to meet challenging **size, weight, and power** constraints

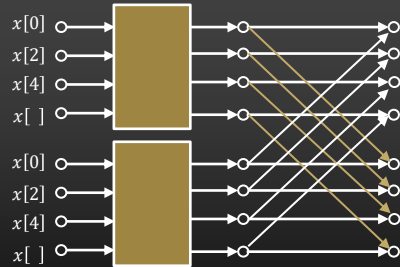


AMD Versal™ RF Series Architecture Overview



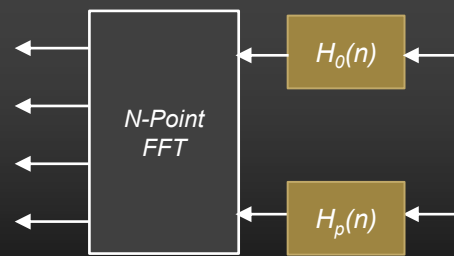
Hard IP for Wideband RF Signal Processing

Increases DSP compute and lowers power consumption



FFT / iFFT

- 4 GSPS, 8-pt to 4K-pt hard IP instance
- Configurable on-the-fly



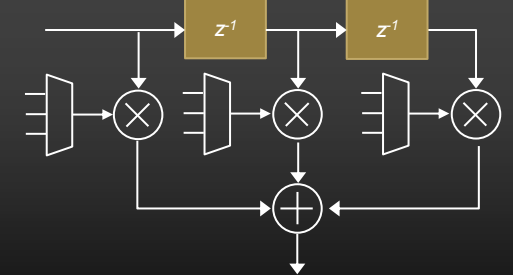
Channelizer

- Separates input signal into N narrow band sub-channels
- Can be used as RX analyzer, TX synthesizer, or a generic filter
- 1st stage in RF-ADC/RF-DAC (channelizer/inverse channelizer or PFIR)
- 2nd stage is connected to and accessible via programmable logic



LDPC Decoder

- Supports 5G, WiFi, and space codes including DVB-S2/S2x



Poly Block¹

- Configurable polyphase filter block
- Interpolating polyphase arbitrary resampler with $(1 \leq R \leq 2)$
- Up to 8 GSPS input and < 1.0 Hz accuracy
- Can be used as a channelizer, polyphase filter & other functions

Multiple Hard IP Instances per Device

AMD Versal™ RF Series Family

Devices tailored to a variety of applications

Hard IP Block		# Blocks VR1602	# Blocks VR1652	# Blocks VR1902	# Blocks VR1952	# Blocks UCle-capable Device (Conceptual)
14-bit RF ADC	# of ADCs	16	4	16	8	NA
	Max Rate (GSPS)	8	32	8	32	NA
14-bit RF DAC	# of DACs	16	8	16	16	NA
	Max Rate (GSPS)	16	16	16	16	NA
Channelizer		224	224	96	96	96
FFT/iFFT		28	28	36	36	36
LDPC Decoder		4	4	6	6	6
AIE Array		126	126	120	120	120
Poly Block ¹		NA	NA	24	24	24
DSP Engines		2,256	2,256	3,976	3,976	3,976
UCle Interfaces	UCle-SP	NA	NA	NA	NA	4x1
	UCle-AP	NA	NA	NA	NA	2x4

1: Each poly block can also be configured as 16 channelizers. A pair of poly blocks can be configured as an arbitrary resampler. See Endnote VER-074



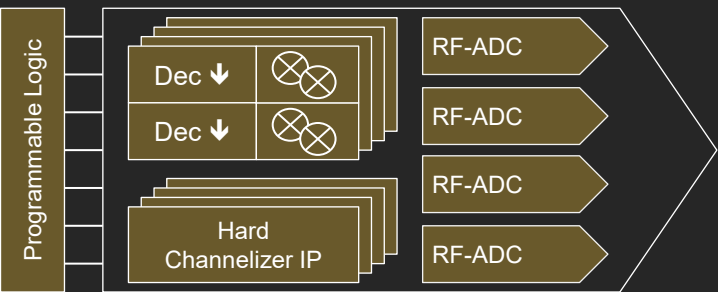
Hard IP Details



AMD Versal™ RF Series Sampling Converters

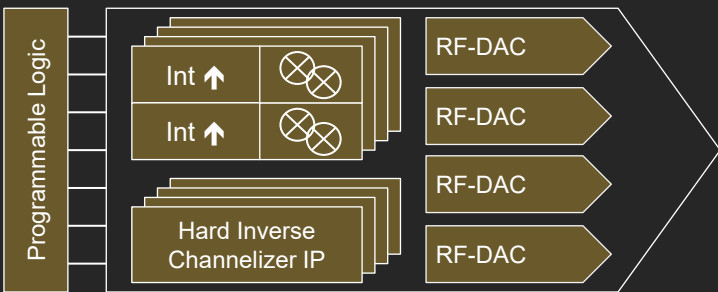
Wide frequency range, high sample rate enables fast and agile signal capture

RF-ADC (Quad) Tile



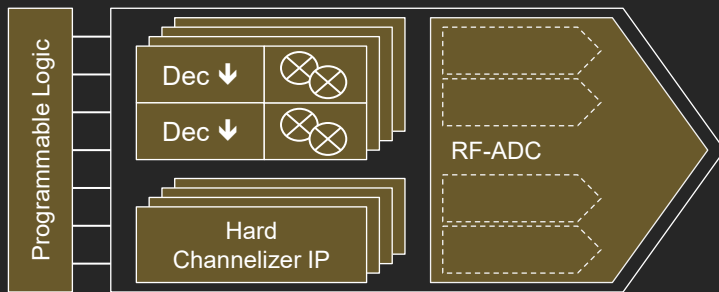
- 4 x 8 GSPS, 14-bit¹
- DC-18 GHz BW @-3 dB

RF-DAC (Quad) Tile



- 4 x 16 GSPS, 14-bit¹
- DC-18 GHz BW @-3 dB

RF-ADC (Single) Tile



- 1 x 32 GSPS, 14-bit¹
- DC-18 GHz BW @-3 dB

Multiple RF-DAC and RF-ADC tiles on each device

- Up to 256 GSPS or 4Tb/s of data from the RF-ADCs and to the RF-DACs

RF-ADC and RF-DAC tiles include following hard IP:

- High Sample Rate DDCs and DUC
 - Not included in processing calculations
 - → Common DSP function in all RF data converters
- 8 GSPS Channelizer / Inverse Channelizer

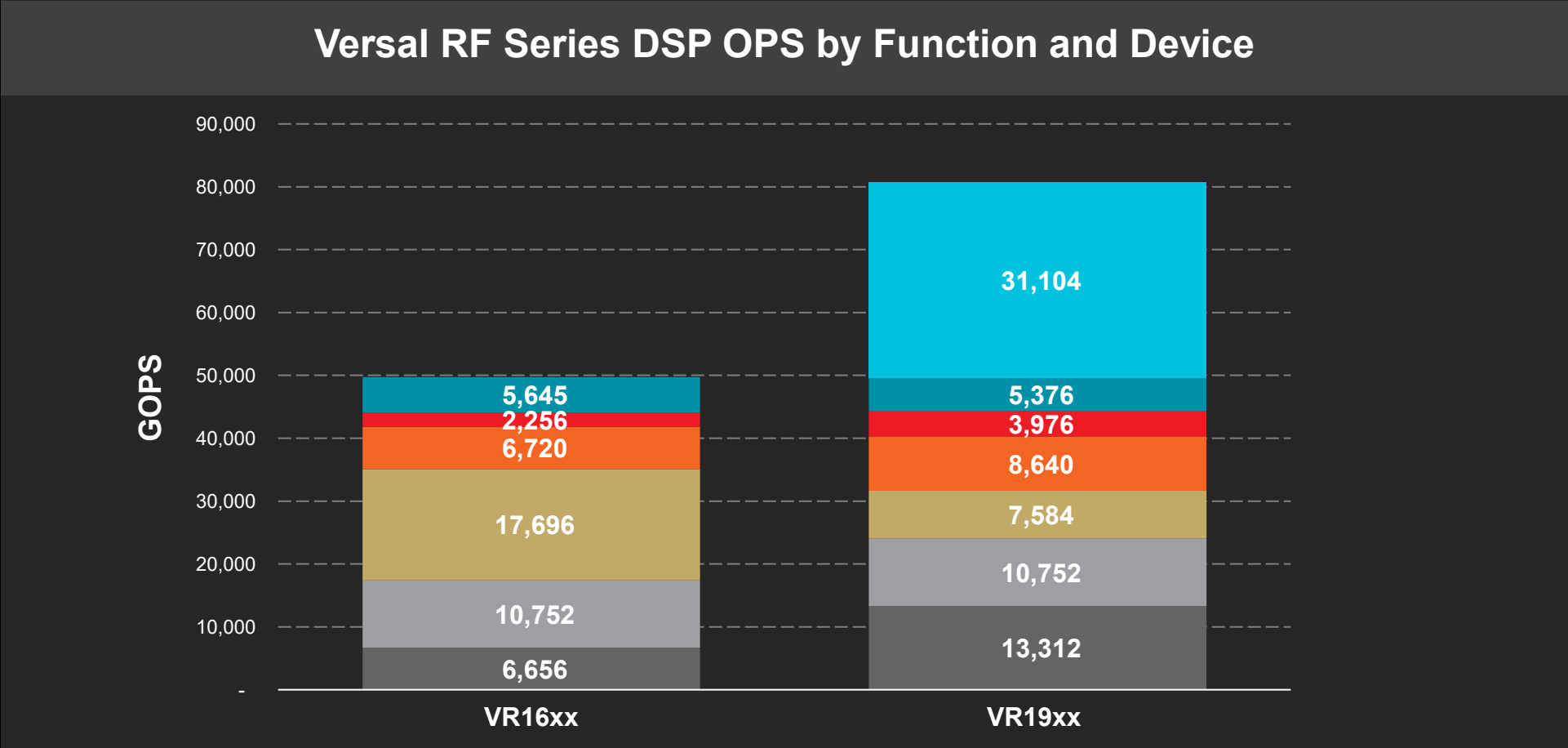
	Device			
	VR1602	VR1652	VR1902	VR1952
16 GSPS RF-DACs	16	8	16	16
8 GSPS RF-ADCs	16	0	16	0
32 GSPS RF-ADCs	0	4	0	8

Total number of available RF-ADCs and RF-DACs vary by device and package

DSP Compute OPs:

What makes the AMD Versal™ RF Series a compelling solution

■ Poly Block ■ AIE Array ■ DSP Engine ■ FFT IP ■ 2nd Stage Channelizer IP ■ RF_DAC 1st Stage Channelizer IP ■ RF-ADC 1st Stage Channelizer IP

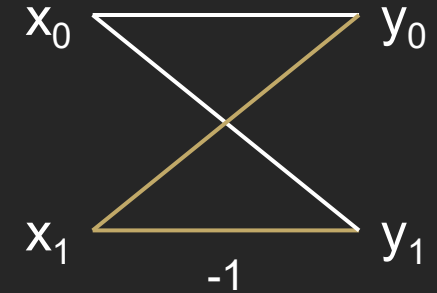


*GOPS case for peak channelizer usage, other use cases of poly block with have different GOPS
See Endnote VER-084

Hard FFT/iFFT Block Details

- Maximum throughput per unit 4 samples/cycle (4 GSPS)
 - VR16xx has 28 4GSPS FFT blocks
 - VR19xx has 36 4GSPS FFT blocks
- Streaming mode
- Power of 2 FFT with point sizes ranging from 8 pt. to 4096 pt.
- Point size & iFFT/FFT change on-the-fly on a block basis
- Input data width 18-bit complex and output 31-bit complex
- Support for scaling and rounding

4K FFT/iFFT



Resource Savings per Instance

Programmable Logic	17.5K LUTs
Flip-Flops	31,344 FF
Block RAM	88 Mb
DSP Engines	320

1.5
Watts

87%

Expected Lower
Power per
Instance

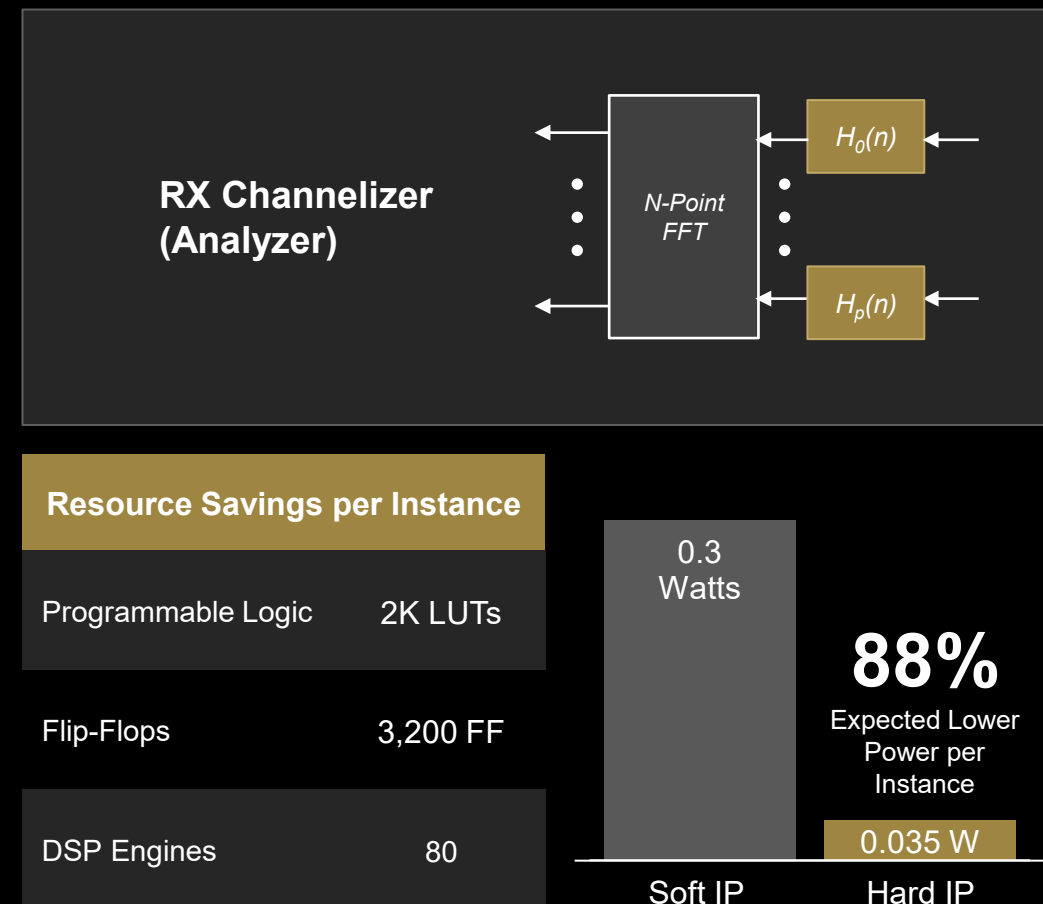
0.203 W

Soft IP

Hard IP

Hard Channelizer Block Details (2nd Stage)

- Full rate programmable 64-tap prototype filter
- 1 GSPS native rate¹
- Critical and 2x oversample modes
- Can be configured as 8 channel polyphase; 8 complex or 16 real taps per channel
- Input can be real in/out: Data rate would be 2 GSPS in slowest grade
- Input can be real in and complex out: Data rate would go down to 1.0 GSPS in slowest grade
- Input can be complex in/out: Data rate would go down to 0.5 GSPS in slowest grade
- 8 pt FFT
- Ability to use as reverse channelizer
- 224 blocks on VR16xx, up to 480 on VR19xx



LDPC Decoder Hard IP Details

LDPC decode only (no LDPC encode or turbo decode)

Enhancements over AMD Zynq™ UltraScale+™ RFSoc SD-FEC

- Support added for DVB-S2/X LDPC codes
- Integrated support for 5GNR (no PL resources required)
- Extended algorithm support; Normalized & Offset Min-Sum
- Programmable input saturation
- Enhanced control options
 - Control I/F or code memory
- Increased parallelism giving higher throughput for large sub-matrix size (aka lift factor size)
 - 7.7 Gb/s @ 8 iterations - 5GNR
 - 4 Gb/s @ 8 iterations - DVB-S2
 - 3.7 Gb/s @ 8 iterations - DVB-S2X

4x LDPC Decoders = 1.3M LUTs Savings

Equivalent resources to a full **AMD Virtex™ UltraScale+™ VU13P FPGA**

LDPC Decode

1		1		1	1	1
		1				1
	1		1		1	
1			1	1		1
	1	1	1		1	
	1		1		1	1

Resource Savings per Instance

Programmable Logic 328K LUTs

Flip-Flops 242,000 FF

Block RAM 784

5.29
Watts

81%

Expected Lower
Power per
Instance

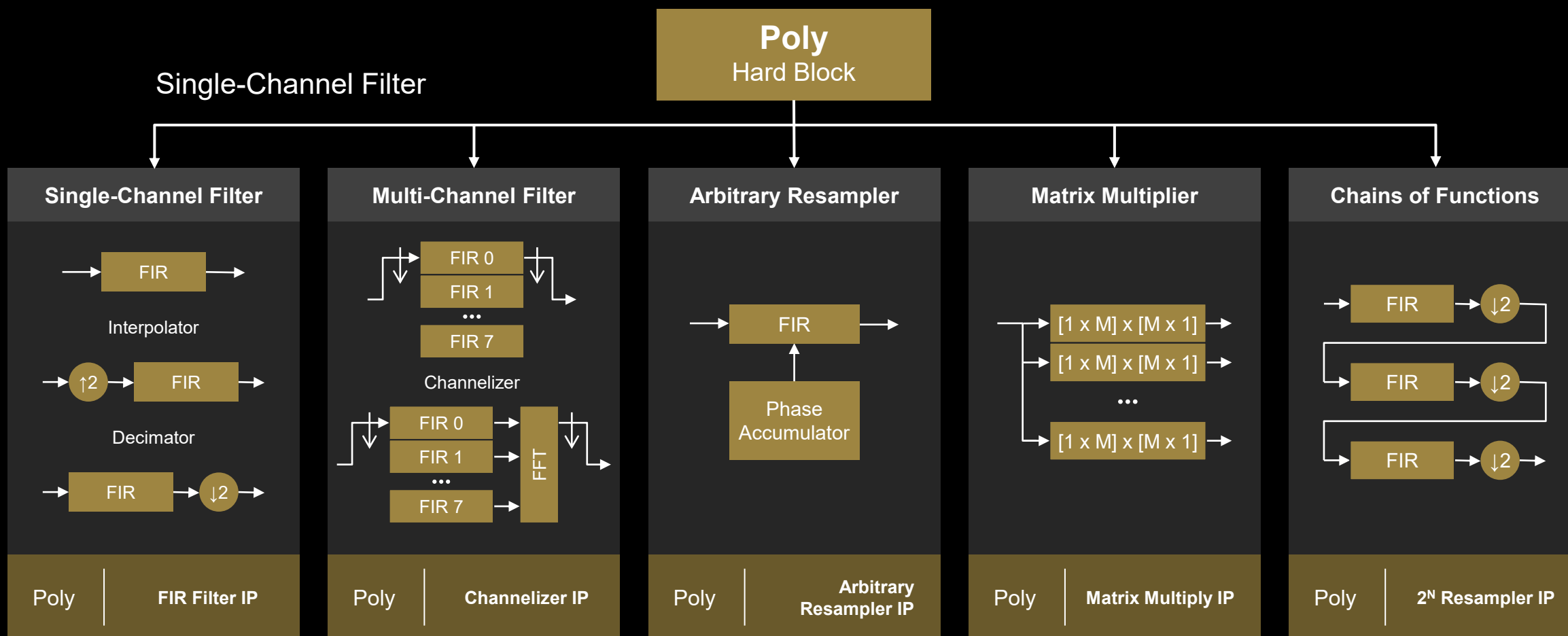
1.02 W

Soft IP

Hard IP

AMD Versal™ Poly Block (VR19xx only)

1 Hard Block, 8 Functions, 5 Soft IPs



AI Engine: Array Architecture

AI Engine array consists of a two-dimensional array of AI Engine tiles

Up to 126 on AMD Versal™ RF Series

Each AI Engine tile contains a very long instruction word (VLIW) processor @ 1 GHz+, integrated memory, and interconnects for streaming, configuration, and debug

DSP Library has configurable kernels to aid development

Software Programmable

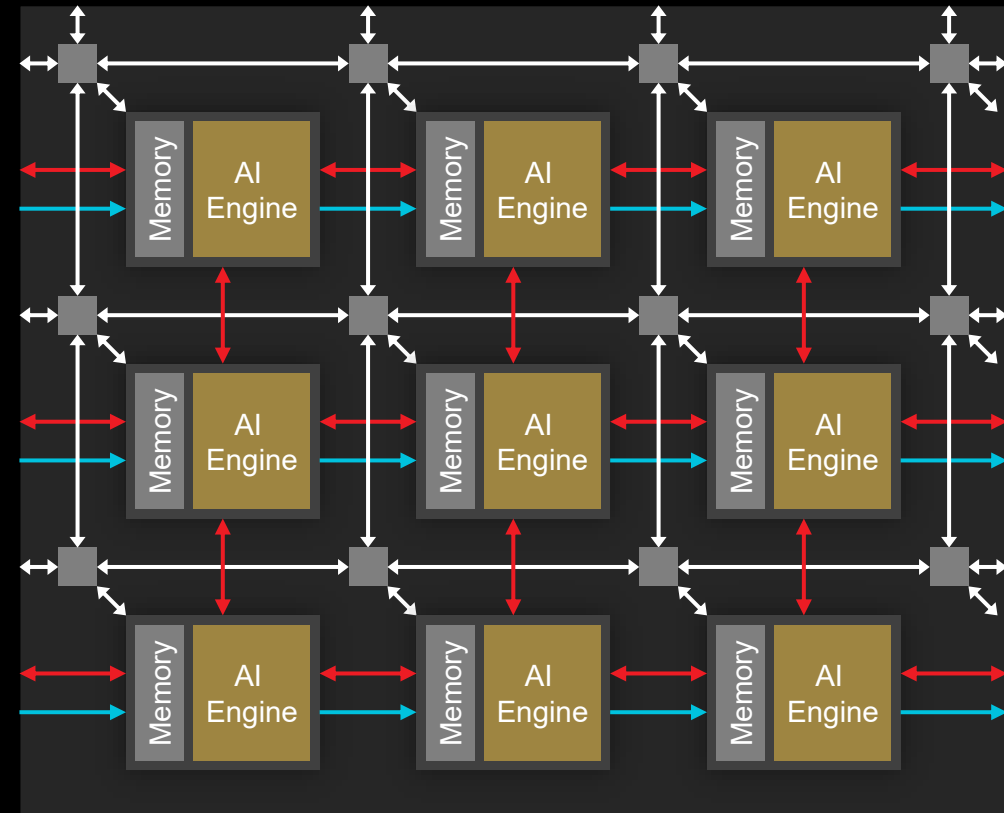
Reduced development time compared to traditional RTL

The VLIW processor runs at 1 GHz+ to implement high-performance DSP functions

Increases DSP compute with 20% less power than traditional Programmable Logic

Array of AI Engines Tiles

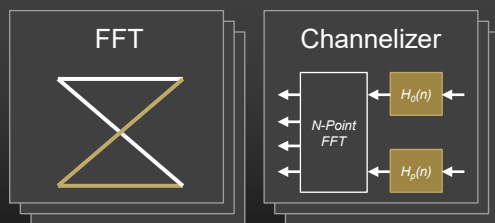
Increase in compute, memory, and communication bandwidth



AI Engines Provide More Compute with Less Power (Example)

32K POINT FFT, 8 GSPS, CINT16 DATA		Internal Data Format: CINT32 FOR AIE; CINT27 FOR PL	Twiddle Factors: CINT16 FOR AIE; CINT24 FOR PL
Metric	PL (540 MHz) 1D FFT Structure	AIE + PL (1250 MHz) 2D FFT Structure	Comments
Resources	171 DSP Engines 153,299 LUTs 54,190 FFs	50 AIEs (16 for compute) 12 DSP Engines 8,052 LUTs (for 16 pt FFT) 6,612 FFs (for 16 pt FFT)	~3.5 DSPs: 1 AIE
UltraRAM / Block RAM	100 block RAM	16 block RAM	Significant block RAM savings
Latency	48 us	7.5 us	
Dynamic Power	9.58W	6.682W 6.138W for AIE .528W for 16 pt FFT	Up to 30% lower dynamic power

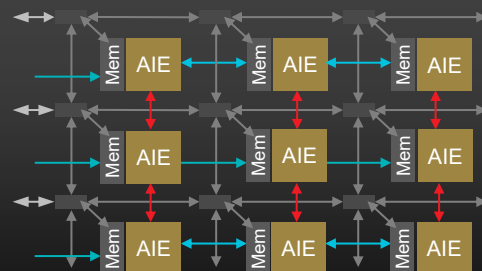
Heterogeneous Compute Types Balance Flexibility and Efficiency



Hard IP¹

Hard Channelizer,² FFT

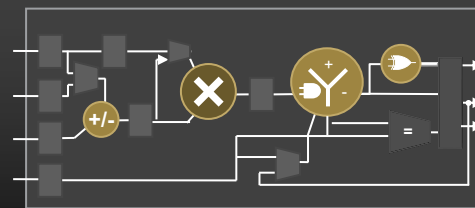
- Enables high sample rates
- Processing for wide bandwidths
- Implements common DSP functions
- Frees up PL for custom functions



AI Engine 2D Array

VLIW and SIMD Architecture

- Efficient matrix functions
- Phased-array operations
- Additional channelization, FFTs, and filters to supplement hard IP



DSP Engines

Hard MULT & Adders

- Custom DSP functions not addressed by AIE or hard IP
- Migrate previous algorithms implemented in DSP Engines



FPGA Fabric

LUT and Memory

- Stitch together hard IP into custom architectures
- Organize data to maximize AI Engine efficiency

Increasing Efficiency

Increasing Flexibility



Solutions



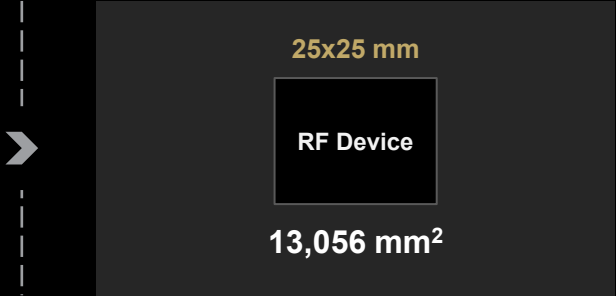
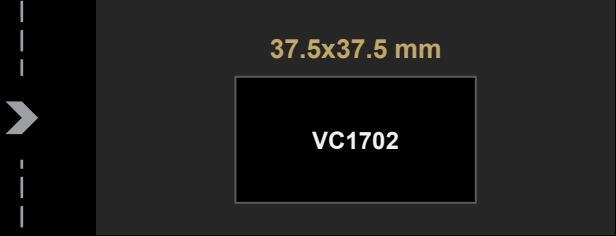
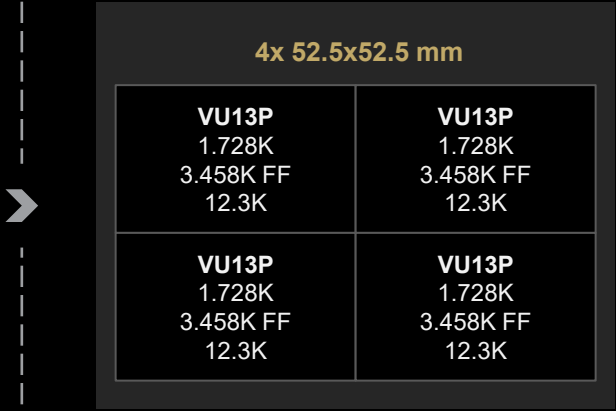
AMD Versal™ RF Size, Weight & Power (SWaP) Advantage

VR1652 Example

DSP Processing and LDPC Decoder			
VR1602/VR1652 Hard IP Equivalent PL Resources	LUTs	Flip-Flops	DSP Engines
FFT Soft	491K	878K	8.96K
1 st Stage Chan. RF-ADC Soft	201K	346K	6.66K
1 st Stage Chan. RF-DAC Soft	269K	431K	10.8K
2 nd Stage Channelizer Soft	455K	717K	17.9K
LDPC Decoder Soft	1,312K	1,008K	-
Total	2,727K	3,380K	44.3K

Adaptive SoC with LUTs, Flip-Flops, DSP Engines, & AI Engine				
VR1602/VR1652 Device Resource	LUTs	Flip-Flops	DSP Engines	AI Engines
Adaptive Logic and DSP Engines	551K	1,102K	2.26K	120

RF Sampling Converter with DUC & DDC				
Versal RF Device	# RF-DACs / RF-ADCs	RF-DAC / RF-ADC Sample-Rate	RF Freq. Range	DUC / DDC
VR1602	16/16	16 / 8 GSPS	DC-18 GHz	Yes
VR1652	4/4	16 / 32 GSPS	DC-18 GHz	Yes



37.5x37.5 mm

VR1602 / VR1652

1,406 mm²

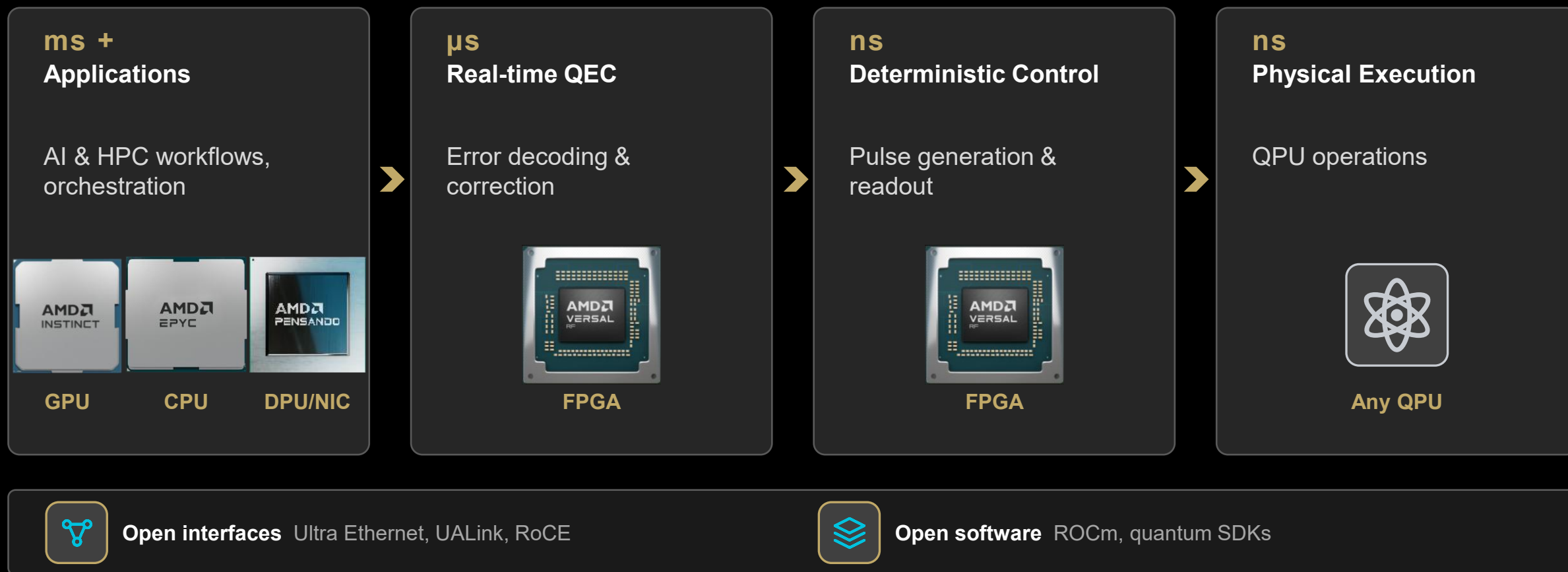
47.5x47.5 mm

VR1602 / VR1652

2,256 mm²

- Equivalent DSP compute of 4x AMD Virtex™ UltraScale+™ VU13P FPGAs (2.7M LUTs, 44K DSP Engines)
- Equivalent PL and AI Engine resources of AMD Versal AI Core VC1702 device (551K LUTs, 120 AIE Tiles)
- RF capability of state-of-the-art discrete data converters

AMD Versal™ RF Series in Quantum Control Applications



Quantum Instrumentation Control Kit (e.g. QICK¹)

QICK with UltraScale+™ RFSoc offers:

- Open qubit control and readout
- Extendable to different qubit platforms
- Built on AMD hardware and software
- AMD RFSoc: Industry's only single-chip CPU + FPGA + Radio Frequency platform
- Open-source AMD PYNQ framework for direct Python and Jupyter Notebook

Versal RF expands that capability offering:

Higher speed IO

- Higher bandwidth → more read channel multiplexing
- Simpler frequency planning

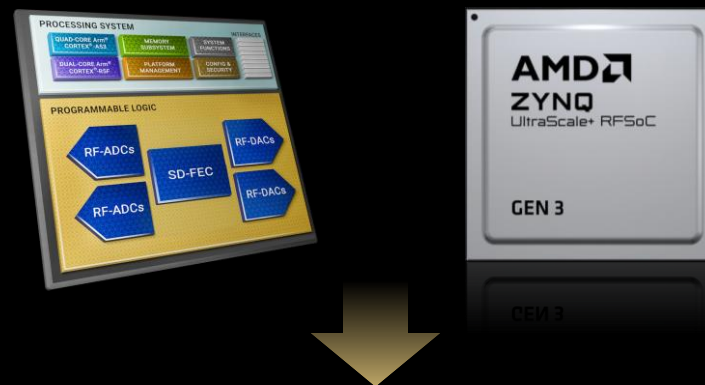
High-speed offload

- Ethernet and/or PCIe5

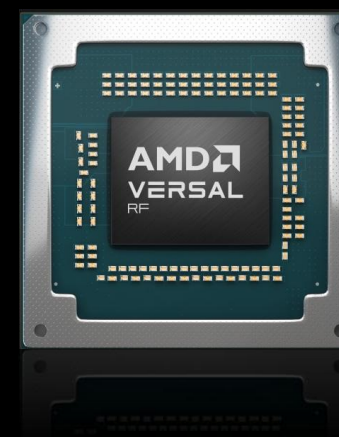
Extensive upgrade in compute and hard IP

- Improved readback
- Enhanced calibration
- QEC
- Near QPU ML

AMD Zynq™ UltraScale+™ RFSoc

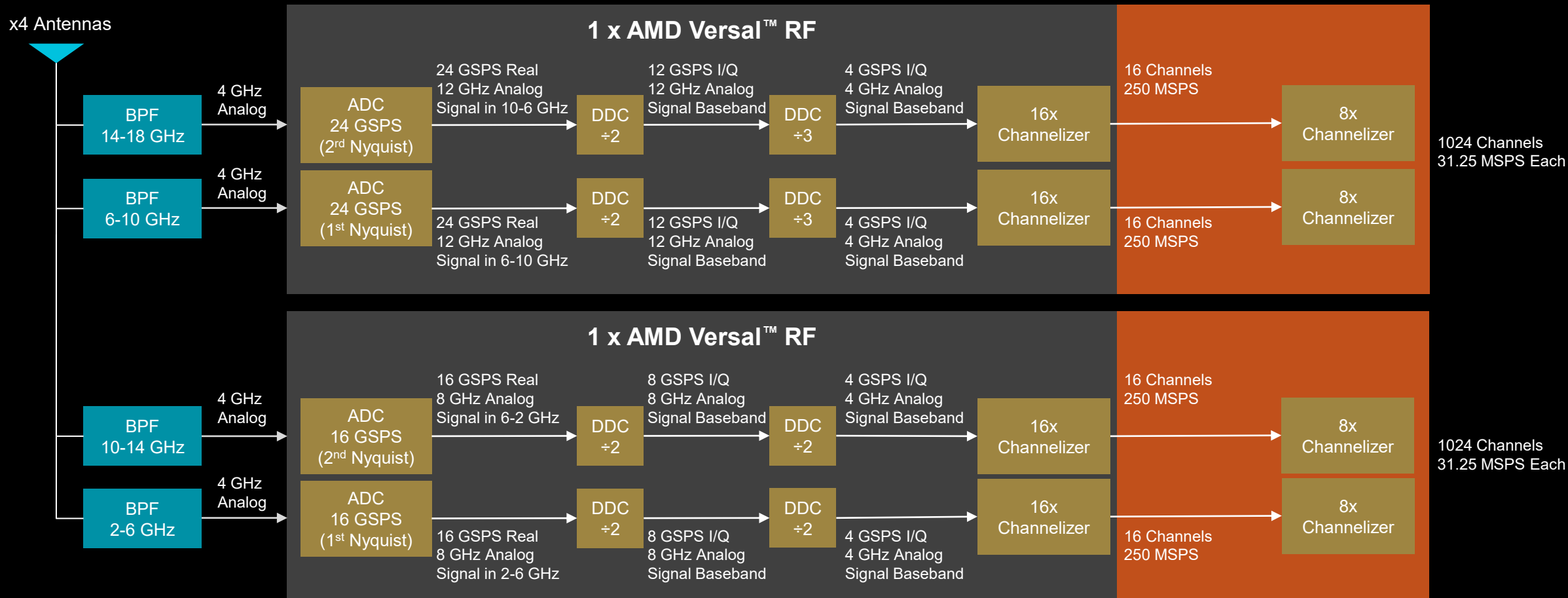


AMD Versal™ RF Series

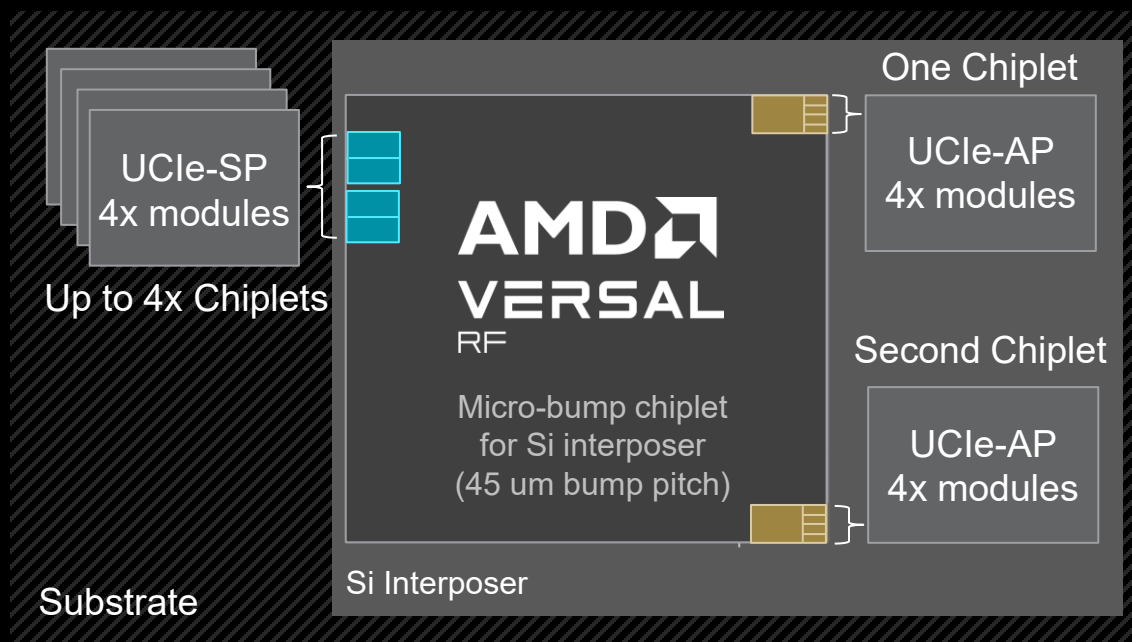
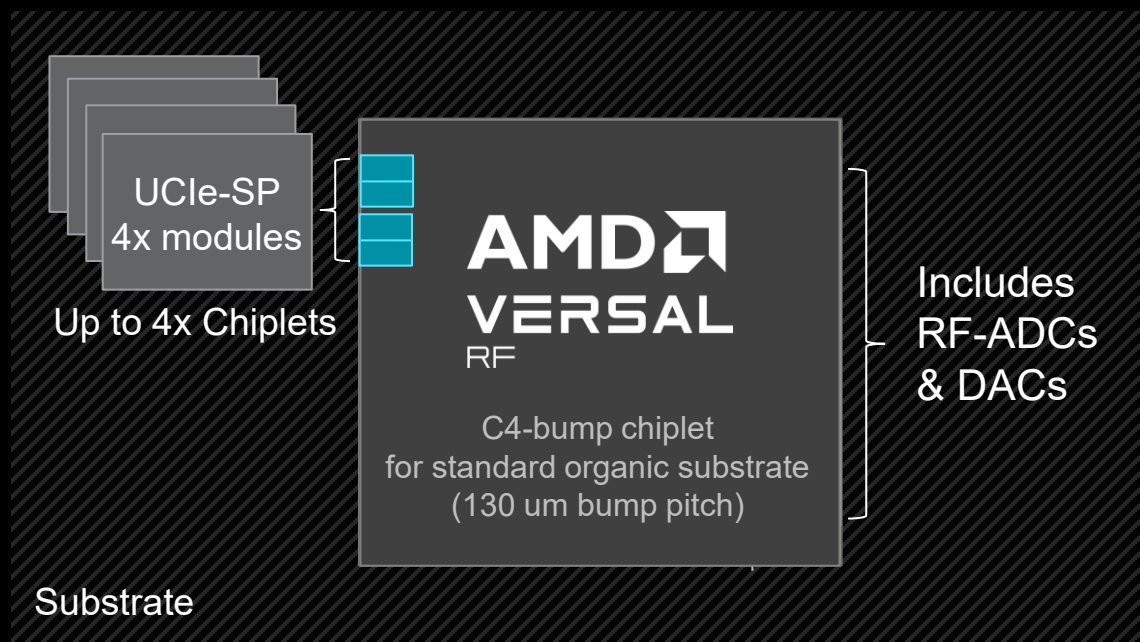


Wideband Spectrum Visibility in AMD Versal™ RF VR1952

■ RF Front End ■ Hard IP ■ Data Converter Tile ■ Programmable Logic



AMD Versal™ RF Series UCle® Chiplet Enablement



UCle v1.1 Connectivity in Select Versal RF Chiplet

- Four independent UCle-SP x16 interfaces, up to 16 GT/s
- Two independent UCle-AP x64 / x32 interfaces, up to 8 GT/s
- Fixed and variable rate support for protocols like JESD204E
- Enabled by AMD extensive experience in known-good die

Enabling Applications Spanning

- Electromagnetic Spectrum Operations (EMSO)
- AI acceleration of analog RF processing
- Test & Measurement
- Communications

Envisioned Applications with Commercial or Custom UCle Chiplets:
RF AFEs, AI, Specialized Compute, CPO / Fiber

AMD Versal™ RF Series



Precise, Wideband Signal Characterization

Wide input frequency, high sample rates for fast and agile spectrum capture



Massive DSP Compute

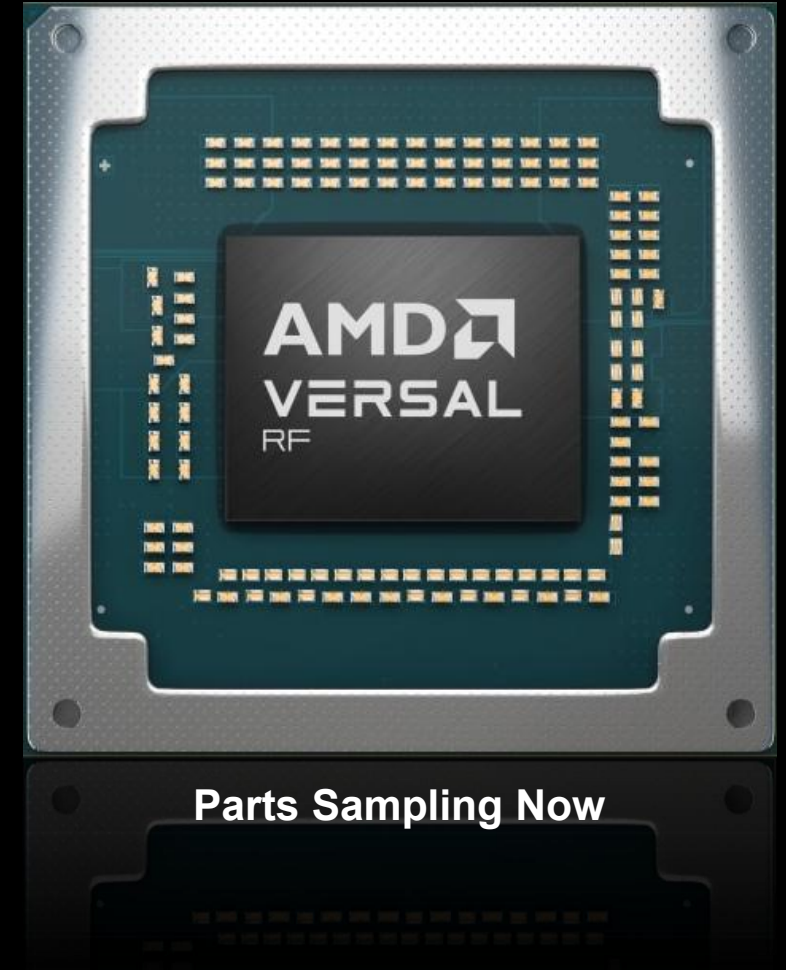
Heterogeneous architecture provides up to 80 TOPS of compute signal analysis



Size, Weight, and Power (SWaP) Optimized

Hard IP & monolithic implementation reduces area & power

Contact Your Account Team for More Information



Parts Sampling Now

Endnotes

VER-037b

Based on 3rd party benchmark testing commissioned by AMD in February 2024, on the AMD Versal adaptive SoC AI Engine and Vitis for AI design tool for a high-precision, high-complexity signal processing application FFT implementation versus traditional programmable logic with Vivado software and Vitis Model Composer tools, version 2023.1. Results will vary based on design specifications.

VER-038

Based on 3rd party benchmark testing, commissioned by AMD in February 2024, on the AMD Versal adaptive SoC with AMD Vitis for AI design tool vs. traditional programmable logic with AMD Vivado software and Vitis Model Composer tools, version 2023.1 in a General Matrix Multiply/GeMM implementation. Results averaged over 2 designs. Results will vary depending on design specifications.

VER-047

Based on 3rd party benchmark testing commissioned by AMD in February 2024, to measure the average dynamic power of the AMD Versal adaptive SoC with AMD Vitis AI design tool versus traditional programmable software implementation with Vivado software and Vitis Model Composer tool, version 2023.1, in a signal processing application FIR implementation. Results will vary depending on design specifications.

VER-067

Based on AMD internal analysis in October 2024, of data sheet specifications for input frequency, max bandwidth, RF-ADC and RF-DAC electrical characteristics, and the maximum RF-ADC and RF-DAC sample rates of the AMD Versal RF devices, compared to the data sheet specifications of the AMD Zynq™ UltraScale+™ RFSoc Gen 3 (prior generation) devices. Actual results may vary based on configuration, device, design, and other factors.

VER-068a

Based on AMD internal analysis (June 2026) to calculate the theoretical DSP compute (including hard IP, AI Engines, and DSP) in channelizer mode expected to be offered by the Versal RF Series devices versus the published DSP compute of the previous-generation Zynq UltraScale+ RFSoc Gen 3 devices. Results and specifications may vary when new AMD products are released in market.

Endnotes

VER-074

Based on an AMD engineering projection of hard IP power values, November 2024. AMD Power Design Manager (2023.2.2) used to determine total power of soft logic, based on Vivado 2023.2.2 IP catalog.

VER-076

Based on an AMD internal analysis, assuming 12 pJ/bit required for JESD204D @ 32 GB/s link, as of November 2024.

VER-077b

Based on AMD engineering projections as of June 2026. Results and specifications subject to change when AMD products are released in market.

VER-080

Based on AMD engineering projections in November 2024, using the hard IP functions in Versal RF Series devices to measure dynamic power versus the total power calculation of an AMD soft logic implementation, as measured with AMD Power Design Manager (2023.2.2), based on AMD Vivado 2023.2.2 IP catalog.

VER-080b

Based on AMD engineering projections in November 2024, using AMD Vivado tools to implement the equivalent hard IP function with programmable logic in a Versal AI Core Series device. Based on Vivado 2023.2.2 IP catalog. Results may vary when products are released in market.

VER-084

Based on AMD internal analysis (June 2026) to calculate the theoretical DSP compute (including hard IP, AI Engines, and DSP) in channelizer mode expected to be offered by the Versal RF Series devices. Actual results will vary when products are released in market.

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